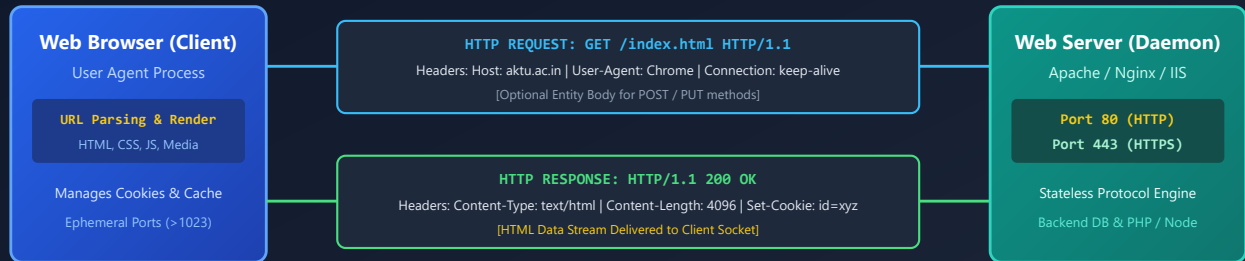


Module 02: World Wide Web & HTTP / HTTPS Protocols

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WORLD WIDE WEB: HTTP & HTTPS TRANSACTION ARCHITECTURE

Request-Response Cycle, Methods, Status Codes, Persistent vs Non-Persistent & SSL/TLS



1. Methods & Status Codes

GET: Fetch resource (idempotent) • **POST:** Submit data / upload payload • **PUT:** Replace resource | **DELETE:** Remove • **200 OK** | **301 Moved** | **304 Not Modified** • **400 Bad Request** | **404 Not Found** • **500 Server Error** | **503 Unavailable**

2. Persistent vs Non-Persistent • **HTTP/1.0 (Non-Persistent):** 1 TCP connection per object (2 RTT delay) • **HTTP/1.1 (Persistent):** Single TCP connection for all webpage objects • **Pipelining:** Send next request without waiting for previous object's ACK

3. HTTP vs HTTPS Security • **HTTP (Port 80):** Plaintext data transfer. Vulnerable to sniffing and MITM attacks. • **HTTPS (Port 443):** HTTP over TLS/SSL. Asymmetric handshake + Symmetric encryption. • **Digital Certificate:** Authenticates server issued by trusted Certificate Authority (CA).

1. HTTP Fundamentals & Stateless Architecture

Hypertext Transfer Protocol is the foundation of data communication for the World Wide Web, operating as a request-response protocol over TCP.

Stateless Character of HTTP

HTTP does not preserve client conversational state between successive requests. To maintain stateful interactions (shopping carts, user logins), modern web architectures use **HTTP Cookies** and **JWT Tokens**.

2. HTTP Methods & Status Code Families

Code Range	Category	Representative Codes
1xx	Informational	100 Continue , 101 Switching Protocols
2xx	Success	200 OK , 201 Created , 204 No Content
3xx	Redirection	301 Moved Permanently , 304 Not Modified (Cache validation)
4xx	Client Errors	400 Bad Request , 401 Unauthorized , 403 Forbidden , 404 Not Found
5xx	Server Errors	500 Internal Server Error , 502 Bad Gateway , 503 Service Unavailable

3. HTTP vs HTTPS Comparison

Parameter	HTTP	HTTPS
Default Port	Port 80	Port 443

Security Transport	Plaintext TCP socket stream	Encrypted over TLS/SSL handshake tunnel
Confidentiality & Integrity	Vulnerable to eavesdropping & MITM attacks	AES symmetric encryption + HMAC integrity verification